IMPLEMENTATION SPECIFICATION

Cacsms Traders

Native NGN Live Trading

Genuine Nigerian Naira accounts on HFM MetaTrader 5. Balance, risk, lot size, margin, and profit and loss are calculated from the connected account. No fixed deposit.

BROKER	CURRENCY	BALANCE	MODES
HFM MT5	**NGN, detected**	**Any account**	**1 and 2 together**

Document control

Field	Value
Platform	Existing Cacsms Traders application and FastAPI backend
Broker	HFM — MetaTrader 5, Zero Spread and other supported types
Account currency	NGN native, alongside USD, EUR, and other reported currencies
Instruments	Existing 28 FX pairs and XAUUSD
Deployment	Current platform plus a dedicated MT5 gateway
Document status	Specification. Live execution is not enabled.
Date	19 October 2026

HOW TO READ THIS DOCUMENT

This is the build specification, not a record of a completed deployment. Live order submission stays off until broker connectivity, contract specifications, order calculations, and safety controls are validated, and the account owner explicitly enables live trading. After that, a change in account balance does not require the system to be rebuilt.

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1. Final implementation requirements

Native Nigerian Naira (NGN) live trading is added to the existing Cacsms Traders platform, using HFM MetaTrader 5. The work integrates into the current architecture. Existing modules stay in place and are not rebuilt or duplicated.

The system supports any account balance. Position size, risk, margin, profit, and loss are calculated from the actual trading account. The capability is permanent and reusable across Demo, Live, and supported Prop Firm accounts.

Component	Implementation
Broker	HFM — MetaTrader 5
Account type	Zero Spread and other supported account types
Base currency	NGN, automatically detected
Starting balance	Dynamic — no fixed amount
Position sizing	Dynamic, based on account equity and risk
Risk management	Configurable percentages and account limits
Profit and loss	Native NGN
Trading instruments	Existing 28 FX pairs + XAUUSD
Trading modes	Mode 1 and Mode 2 simultaneously
Order execution	Live MT5 trading supported
Account synchronization	Real-time broker balance, equity, positions, and margin
Deployment	Existing Cacsms Traders + dedicated MT5 gateway

STANDING RULE

Deposit amounts, account balances, exchange rates, and account-specific leverage are never hardcoded. Every calculation uses the values the broker reports for that account.

2. Live trading architecture

Market data, account state, and orders pass through one secure MT5 gateway. Cacsms Traders keeps analysis and scoring. Both autonomous modes submit candidates to a single NGN risk and execution engine.

BROKER ACCOUNT

HFM MT5 live account — NGN

Broker balances · Market quotes · Contract specifications

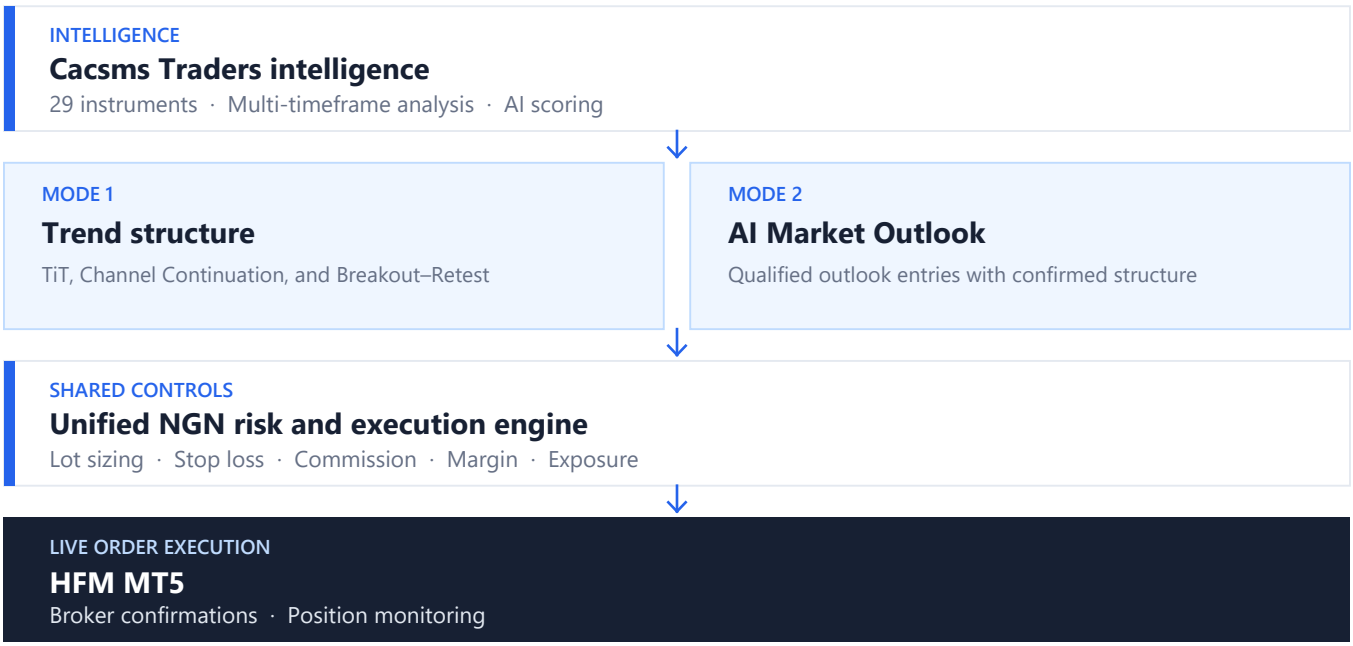
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GATEWAY

Secure MT5 gateway

Connection · Synchronization · Orders · Reconciliation

↓



ONE SHARED CONTROL PLANE

Mode 1 and Mode 2 may operate at the same time. They share one centralized risk budget, one account exposure limit, and one execution lock. That lock prevents conflicting or duplicate orders.

3. Detailed implementation

The ten workstreams below are the production scope. Each one extends the current Cacsms Traders platform. None of them introduces a second copy of an existing module.

3.1 Dynamic account management

The platform detects the connected broker account and keeps it synchronized. NGN is a native account denomination, together with USD, EUR, and any other currency the broker reports.

- Synchronize account currency, balance, equity, leverage, margin, free margin, margin level, floating profit and loss, and trading permissions.
- Use actual broker-reported values for every calculation. Never hardcode deposit amounts, account balances, exchange rates, or account-specific leverage.
- Support multiple tenants and multiple trading accounts, including Demo, Live, and supported Prop Firm accounts.

3.2 HFM MT5 live connection

The existing MT5 integration is extended with a dedicated, secure, persistent gateway on a compatible Windows VPS or server. The MetaTrader 5 desktop terminal is not run inside Vercel serverless functions.

- Authenticate communication with the FastAPI backend and store credentials encrypted.
- Monitor connection health and reconnect automatically.
- Discover symbols, stream market quotes, retrieve historical candles, and load broker contract specifications.
- Place, modify, and close orders, and synchronize trade history.
- Use the account's actual HFM symbol names and trading conditions. Never assume a fixed symbol suffix.

3.3 Native NGN accounting

Accounting throughout Cacsms Traders is currency-aware. Amounts are shown in the account's actual currency, including the naira sign (₦) for NGN accounts.

- Display balance, equity, available margin, floating profit and loss, realized profit and loss, commission, swaps, deposits, withdrawals, and drawdown in the account currency.
- Use Decimal or equivalent fixed-precision arithmetic for financial records.
- Store currency codes, conversion rates, conversion timestamps, broker transaction references, and reconciliation status.
- When an instrument's profit currency differs from the account currency, use validated broker profit calculations or a correctly converted tick value. Never treat a USD pip value as an NGN value.

3.4 Dynamic position sizing

Volume is derived from current equity, the configured risk percentage, the protective stop, and the broker's contract specification. The sizer does not raise risk in order to force a trade.

- Calculate the permissible risk amount from current broker equity and the configured risk percentage.
- Derive volume from entry price, stop-loss price, broker contract specifications, account-currency profit calculations, and expected transaction costs.
- Respect minimum lot size, maximum lot size, lot step, margin requirements, stop levels, and execution restrictions.
- Round volume down to the permitted increment and recalculate the projected loss.
- Reject the trade when the minimum permitted volume exceeds the risk budget. Never increase risk automatically to force execution.

3.5 Autonomous trading modes

Both existing modes remain, and they may run together. Candidates from either mode enter the same risk engine and the same order queue.

Mode	Strategies	Submission
Mode 1	Trend-in-Trend, Channel Continuation, Breakout–Retest	Central risk engine and execution queue
Mode 2	AI Market Outlook	Central risk engine and execution queue

- Keep the existing multi-timeframe analysis, channel detection, fractals, strength matrix, AI scoring, and confirmed-entry requirements.
- Apply the existing opportunity-confidence threshold, and also require independent technical entry confirmation, acceptable trading costs, sufficient margin, and a valid protective stop-loss.
- Prioritize XAUUSD analysis. Do not force execution because the instrument is prioritized.

3.6 Live order execution

Live orders follow one controlled pipeline. An uncertain broker response is never retried blindly. Broker orders and positions are reconciled first.

1. Opportunity detected
2. Entry confirmed
3. Risk calculated
4. Volume validated
5. Margin checked

6. Account exposure checked
7. Broker order check
8. Live order submission
9. Broker confirmation
10. Position monitoring
11. Trade reconciliation

- Carry an idempotency key on every trade request, and hold an execution lock for the duration of submission.
- Verify the broker result. Use a bounded retry policy only when the outcome is known to be safe to repeat.
- Prevent duplicate orders, and write a persistent execution audit log for every attempt and result.

3.7 Risk and capital protection

Limits are configured as percentages and account-level caps. They are evaluated against current equity and recorded equity baselines, not against a fixed initial deposit. Exposure is aggregated across both autonomous modes. New entries are blocked when a limit is exceeded.

Control	Rule
Per-trade risk	Configurable percentage of current broker equity
Concurrent positions	Configurable maximum number of open positions
Aggregate exposure	One cap shared by Mode 1 and Mode 2
Daily loss	Configurable limit; blocks new entries when reached
Weekly loss	Configurable limit; blocks new entries when reached
Maximum drawdown	Measured from recorded equity baselines
Spread threshold	Reject entries when the spread is outside the configured bound
Margin protection	Require sufficient free margin before submission
Emergency stop	Manual suspension, recorded in the audit log
Live-trading switch	Auditable enable and disable. Default is disabled.

3.8 Frontend integration

Existing screens are extended. The professional white-background design, modern icons, and responsive layouts stay as they are.

- Extend Account Management, Cacsms Engine, Market Intelligence, Opportunities & Risk, Connections, Trade Execution, and Performance Reporting.
- Show native NGN values with ₦ formatting, synchronized account status, live-trading permission, equity, margin, available risk, open positions, active strategies, execution events, and the reason a trade was rejected.
- Distinguish connected, disconnected, degraded, simulation, and live-execution states so an operator can see which state the account is in.

3.9 Database and API

Migrations are backward-compatible for SQLite in development and PostgreSQL in production. Every account operation is authenticated and isolated by tenant.

- Extend broker accounts, trading permissions, risk policies, currency metadata, market quotes, conversion snapshots, trade requests, execution results, positions, transaction costs, equity baselines, and audit logs.
- Provide authenticated REST and WebSocket endpoints for broker synchronization, risk evaluation, live execution controls, position updates, and account reporting.
- Enforce strict tenant isolation and authorization on all account operations.

3.10 Production deployment and testing

The MT5 gateway is deployed on a compatible persistent host and connected securely to the existing FastAPI backend. The implementation builds live execution, and leaves it disabled until the checks below have passed and the account owner turns it on.

- Unit tests for currency-aware accounting and risk limits.
- Broker-adapter integration tests, including symbol discovery without a fixed suffix.
- Simulated execution tests and order-reconciliation tests, including an uncertain broker response.
- NGN position-sizing tests, including a minimum lot that exceeds the risk budget.
- Multi-mode concurrency tests for the shared risk budget, exposure limit, and execution lock.
- Account-disconnection and recovery tests, including automatic reconnection.
- Deliver integrated frontend and backend code, database migrations, configuration templates, automated tests, deployment instructions, and operational documentation.
- Preserve all existing Cacsms Traders functionality.

GATE BEFORE LIVE ORDERS

Trading stays disabled until broker connectivity, contract specifications, order calculations, and safety controls have been validated, and the account owner explicitly enables live trading.

4. Live trading activation

The implementation supports three operational states. Connection and accounting can be verified while live order submission remains off.

State	Capability	Status
Broker connected	Market data and account synchronization	Verification required
NGN risk engine	Dynamic volume and loss calculations	To implement
Live execution	Real broker order submission	Not yet enabled

VERIFICATION REQUIRED

Broker connected means market data and account synchronization are in place and still need to be confirmed against the live HFM account.

TO IMPLEMENT

The NGN risk engine calculates dynamic volume and projected loss from current equity, the stop, contract specifications, and transaction costs.

NOT YET ENABLED

Live execution is the only state that submits real orders. It is specified here and is not switched on by this document.

5. Implementation decision

Cacsms Traders will support genuine NGN-denominated HFM MT5 accounts, with no fixed deposit requirement. The trading amount, risk budget, lot size, and margin are determined dynamically from the connected account and from current broker conditions.

Once this specification is implemented and verified, live trading can be enabled on the existing HFM account. A later change in account balance does not require the system to be rebuilt.

Decision	Consequence
Native NGN	Profit, loss, margin, and equity use the account currency, including ₦ formatting.
Dynamic balance	Any deposit size is supported. Nothing is tied to a starter amount.
Shared risk	Mode 1 and Mode 2 draw on one budget, one exposure limit, and one execution lock.
Live stays off	Order submission waits for validation and an explicit enable by the account owner.
Existing platform	The current Cacsms Traders product is extended. It is not replaced.